ModelIt!

Teacher's Guide

De-Extinction: Should We Bring Back the Woolly Mammoth?

Modeling the Genetics, Ecology, and Ethics of Resurrecting Extinct Species

Quick Reference

Grade Level	Grade 10L3
NGSS Standard	HS-LS4-4, HS-LS2-7
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS4-4, HS-LS2-7

Construct an explanation based on evidence for how natural selection leads to adaptation of populations. Design, evaluate, and refine a solution for reducing the impacts of human activities on the environment and biodiversity.

Three-Dimensional Learning

Science Practice	**Constructing Explanations and Designing Solutions** — Students construct evidence-based explanations of why natural selection shaped mammoth adaptations and design solutions for the genetic, ecological, and ethical challenges of de-extinction.
Disciplinary Core Idea	**LS4.C Adaptation / LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Students analyze how natural selection created mammoth adaptations to Pleistocene conditions and evaluate whether modern ecosystems can support species adapted to environments that no longer exist.
Crosscutting Concept	**Stability and Change** — Students model the tension between ecosystem stability (which has reorganized without mammoths) and deliberate change (reintroducing a species to restore ecosystem functions), evaluating whether change would create new stability or new disruption.

CAST Testing Alignment

MULTIPLE CHOICE:

A de-extinction team has successfully created a healthy mammoth-elephant hybrid calf using CRISPR gene editing and surrogate pregnancy. According to your model, which variable represents the NEXT most critical barrier to establishing a self-sustaining wild population?

CONSTRUCTED RESPONSE:

Using your model, argue whether the woolly mammoth de-extinction project should proceed, be modified, or be redirected to protect existing endangered species. Address at least five of the ten model variables in your evidence-based argument.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

In 2021, Colossal Biosciences received $15 million in funding to bring back the woolly mammoth — or at least a cold-adapted elephant with mammoth traits. The last mammoths died approximately 4,000 years ago on Wrangel Island, and well-preserved specimens in Siberian permafrost have yielded significant amounts of ancient DNA. But the distance between recovering DNA fragments and restoring a self-sustaining wild population spans ten interconnected challenges that together define the frontier of both biotechnology and conservation biology.

The Ancient DNA Challenge

Mammoth DNA recovered from permafrost specimens is always fragmented — broken into millions of short pieces averaging 50-200 base pairs long (the full genome is 4 billion base pairs). Fragments must be painstakingly assembled, like completing a puzzle where most pieces are missing and many are damaged. Chemical modifications from centuries of degradation (cytosine deamination, strand breaks, cross-linking) create systematic errors. No complete mammoth genome has been assembled; current references are about 80-90% complete, with gaps filled using the Asian elephant genome as a template.

From Genome to Organism

Even with a complete genome, creating a living mammoth requires multiple additional steps. CRISPR gene editing can modify Asian elephant cells with mammoth-specific genes (cold adaptation, hemoglobin variants, fat distribution, hair growth). But off-target edits — unintended changes elsewhere in the genome — can cause developmental abnormalities. The modified cells must be used to create embryos through somatic cell nuclear transfer (cloning). The embryo must develop in a surrogate elephant mother for 22 months — the longest gestation of any land mammal. Success rates for cloning in mammals are typically 1-5%.

The Ecological Gauntlet

A healthy mammoth born today would face an environment radically different from the Pleistocene. The mammoth steppe — a vast grassland spanning Eurasia and North America — has largely transitioned to boreal forest and tundra. Food plants have changed. New diseases, parasites, and competitors have evolved. Predators the mammoth never encountered (including modern humans with firearms) now exist. The animal would have no cultural knowledge of migration routes, seasonal food sources, or social behaviors that were transmitted between generations in the original population. It would be genetically a mammoth but ecologically a stranger.

The Ethics of Resurrection

De-extinction raises profound ethical questions. Should resources be spent resurrecting extinct species when thousands of existing species face imminent extinction? What is the welfare status of a mammoth born into a world it was not designed for — is it ethical to create an animal that may suffer? If the de-extinct population requires permanent human management, is it a wild species or an elaborate zoo exhibit? And philosophically, is a mammoth-elephant hybrid with 99% elephant genes actually a mammoth, or something entirely new? These questions do not have scientific answers — they require ethical frameworks that society has not yet developed.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the de-extinction system: DNA recovery quality, gene editing accuracy, surrogate compatibility, population viability, ecosystem readiness, habitat availability, genetic diversity, disease resistance, ecological integration, and ethical threshold.

E **Establish Relationships**

Students map relationships showing that DNA recovery quality and gene editing accuracy determine the starting biological material, which must survive surrogate gestation, then face ecological barriers (ecosystem readiness, habitat availability, disease resistance) that determine whether a viable population can establish, all within the constraints of ethical societal acceptance.

V **Visualize & Model**

Students build a computational model showing the ten-component de-extinction system with cascading barriers from genome to ecosystem.

E **Evaluate Outcomes**

Students run best case technology, ecological mismatch, and ethics versus science scenarios to identify which barriers are most limiting and whether any configuration produces a viable outcome.

R **Revise & Extend**

Students add behavioral repertoire, climate trajectory, or keystone effects to model more realistic long-term de-extinction outcomes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"4,000 years ago, the last woolly mammoth died on a remote Arctic island. Its DNA has been recovered from the permafrost. We have the gene editing technology to create a mammoth-like animal. The question is not CAN we — it is SHOULD we?"*

Watch For

Show the Wrangel Island mammoth story. Reveal that Colossal Biosciences is actually trying to do this right now with real funding. Ask: Who is excited? Who is concerned? Both responses are valid.

Slide 2: Learning Objectives

3 min

Say: *"Today you are evaluating the most ambitious biology project ever proposed — resurrecting an extinct species. Your model will determine whether it is scientifically viable, ecologically wise, and ethically justified."*

Watch For

Have students read objectives. Pre-teach 'ancient DNA' and 'minimum viable population.' Emphasize that this lesson integrates genetics, ecology, AND ethics — all three matter.

Slide 3: Big Question

3 min

Say: *"The technology is approaching feasibility. But should we do everything we CAN do? That is not a science question — it is a science AND ethics question."*

Watch For

Quick-write: Students take an initial position — should the mammoth de-extinction project proceed? Record positions to revisit after the model exploration. Emphasize evidence-based reasoning.

Slide 4: LEVER Framework

2 min

Say: *"We are building a model with ten variables spanning molecular genetics, reproductive biology, ecology, conservation, and ethics. De-extinction is the ultimate interdisciplinary challenge."*

Watch For

Preview LEVER steps. Emphasize that this model includes a non-scientific variable (Ethical Threshold) because real-world scientific projects do not exist in an ethical vacuum.

Slide 5: Activity 1: Components

8 min

Say: *"Ten variables. Three you can influence through technology and societal decisions. Seven that biology and ecology determine. Can you get all ten to viable levels simultaneously?"*

Watch For

Guide students through all ten components. Spend extra time on the transition from genetics (first three) to ecology (next five) to ethics (last two) — the challenge shifts in nature at each stage.

Slide 6: Activity 2: Connections

8 min

Say: *"Perfect DNA recovery does not help if the ecosystem cannot support the animal. Perfect ecology does not matter if gene editing accuracy is too low. Every variable must clear its threshold or the whole project fails."*

Watch For

Help students map the cascading dependencies. Show how genetic barriers must be cleared before ecological barriers become relevant, and both before ethical questions can be fully evaluated.

Slide 7: Activity 3: Simulation

10 min

Say: *"Can a mammoth actually survive in the modern world? Let us test the best technology against the reality of today's ecosystems."*

Watch For

Students predict outcomes for each scenario. Run simulations. The ecological mismatch scenario should reveal that solving the genetics is the easy part — the ecology is the real barrier.

Slide 8: Discoveries

5 min

Say: *"De-extinction is not about bringing back one animal — it is about restoring an ecologically functional population in a world that has moved on. The mammoth is a symbol, but the real question is about ecosystem restoration."*

Watch For

Revisit initial positions. How has the model changed perspectives? Discuss the permafrost preservation argument — if mammoths could slow Arctic carbon release, does that change the ethics?

Slide 9: STEM Challenge

12 min

Say: *"You are the independent scientific review panel. $150 million is on the table. Proceed, modify, or redirect? Your recommendation must be based on evidence from all ten model variables."*

Watch For

Groups develop comprehensive viability assessments. Present recommendations with evidence from each of the ten variables. Class debates the different conclusions — there is no single correct answer.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** DNA Recovery Quality, Gene Editing Accuracy, Ethical Threshold
- **INTERNAL:** Surrogate Compatibility, Population Viability, Ecosystem Readiness, Habitat Availability, Genetic Diversity, Disease Resistance, Ecological Integration

Relationships

- **DNA Recovery Quality to Gene Editing Accuracy:** POSITIVE (+) — Higher quality ancient DNA — longer fragments, fewer damage artifacts, more complete genome coverage — reduces the number and complexity of edits needed, making the gene editing process more accurate and reducing the risk of introducing errors or missing critical sequences.
- **Ecosystem Readiness to Population Viability:** POSITIVE (+) — Higher ecosystem readiness — appropriate food sources, suitable climate, manageable predation pressure, absence of novel pathogens — increases the survival and reproductive success of reintroduced individuals, enabling the population to grow toward self-sustaining numbers.
- **Genetic Diversity to Disease Resistance:** POSITIVE (+) — Greater genetic diversity in the founding population provides a broader immune repertoire, increasing the probability that at least some individuals carry resistance to novel pathogens encountered in the modern environment. Low diversity populations are devastatingly vulnerable to disease epidemics.

Key Discoveries

- De-extinction is not a single technological challenge but a cascade of ten interdependent barriers — solving the genetics is necessary but insufficient without solving ecology, health, and ethics simultaneously
- Ancient DNA is always incomplete and damaged — the 'mammoth' genome is actually a patchwork of recovered mammoth sequences and gap-filling from Asian elephant DNA, creating a hybrid rather than a true resurrection
- The ecological niche a species occupied may no longer exist — the mammoth steppe ecosystem has largely disappeared, and reintroducing mammoths would require either habitat restoration or accepting that the animals will live in a fundamentally different environment
- The minimum viable population problem means de-extinction is not about creating one animal but about creating hundreds to thousands with sufficient genetic diversity to avoid inbreeding collapse

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"De-extinction means bringing back the exact same species"	Current de-extinction technology creates a hybrid — not a resurrection. The mammoth genome is assembled from incomplete ancient DNA with gaps filled by Asian elephant sequences. CRISPR edits modify elephant cells with mammoth-specific genes, but the resulting organism is an elephant with mammoth traits, not a true mammoth. Its mitochondrial DNA, epigenetic programming, microbiome, and developmental environment are all elephant. Whether this constitutes a mammoth is a philosophical question, not a scientific one. The model treats this as a variable: what percentage of the genome must be mammoth for the organism to count?

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"If we can clone a mammoth, the species is saved"	Creating one or even ten individuals is not saving a species. A species requires a self-sustaining population with enough genetic diversity to avoid inbreeding depression over generations. The minimum viable population for most large mammals is 500-5,000 individuals. Creating this many mammoths would require thousands of cloning attempts, each with a 1-5% success rate, using surrogate elephants with 22-month gestations. The timeline to reach viable population size is decades to centuries, and every individual in that population must survive in the wild. A few animals in a zoo is not species restoration.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"De-extinct animals can just be released into the wild"	Release into the wild requires suitable habitat (which may have changed dramatically), appropriate food sources (which may have shifted), absence of novel threats (new diseases, predators, human activity), and learned behaviors (migration routes, social structures, foraging strategies) that cannot be genetically inherited. Most de-extinct animals would initially require managed reserves — essentially large-scale captive habitats — with gradual introduction to more natural conditions. Even wild-born animals of existing species raised in captivity often fail to survive release. A de-extinct animal facing a world it never evolved in would face even greater challenges.